A Genealogical Theorem for Prime-Generating Monic Polynomials

with Examples in Degrees 3 and 4

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Abstract

We formulate and prove a genealogical theorem for prime-generating monic integer polynomials evaluated on the non-negative integers. The key result states that any such polynomial with run length $L > 1$ admits a unique parent with run $L-1$ under a canonical shift of the argument. This induces rooted genealogical trees whose leaves correspond to structural maxima for the prime run length. We provide full formal proofs, structural corollaries, and computational examples, including a monic quartic with run $L = 38$ and a monic cubic with run $L = 31$ in the canonical setting.

1 Introduction

Prime-generating polynomials have long attracted interest in number theory and experimental mathematics. A classical example is Euler's quadratic

$$E(n) = n^2 + n + 41,$$

which produces prime values for $n = 0, 1, \dots, 39$, hence has a prime run of length 40 on the non-negative integers. In higher degree, and in particular for monic cubic and quartic polynomials, long prime runs are considerably rarer and less well understood.

In this paper we focus on *monic* integer polynomials

$$f(n) = n^d + a_{d-1}n^{d-1} + \cdots + a_1 n + a_0 \in \mathbb{Z}[n], \quad d \geq 2,$$

and on their prime-generating behavior on the domain $n \geq 0$. We adopt the standard convention that primality is tested on the absolute value $|f(n)|$, but in the *canonical* setting we are interested in polynomials that actually take positive values on the relevant interval, so that $|f(n)| = f(n)$.

The central contribution of this note is a *genealogical theorem* for such polynomials: any canonical monic polynomial that generates a non-trivial run of primes ($L > 1$) on $n = 0, 1, \dots$ is the *child* of a unique canonical monic polynomial with run length exactly $L-1$. The parent is obtained by a simple shift of the argument. Iterating this construction yields finite genealogical chains descending from a leaf (often a record polynomial) down to a root with $L = 1$.

This view organizes prime-generating polynomials into rooted trees. We show that record examples arise naturally as leaves or near-leaves of these trees and illustrate the framework with two computational case studies: a monic quartic with run $L = 38$ and a monic cubic with run $L = 31$ in the canonical monic category.

2 Definitions and framework

We begin with the basic notions used throughout.

Definition 2.1 (Run length)**.** Let $f(n) \in \mathbb{Z}[n]$. The *prime run length* $L(f)$ of f (on the non-negative integers) is defined by

$$L(f) := \max\Big\{k \in \mathbb{Z}_{\geq 0} \;\Big|\; |f(n)| \text{ is prime for all } n = 0, 1, \dots, k-1\Big\}.$$

If $|f(0)|$ is not prime, we declare $L(f) = 0$.

Definition 2.2 (Canonical monic polynomial)**.** A polynomial $f(n) \in \mathbb{Z}[n]$ is called *canonical monic of degree* d if

$$f(n) = n^d + a_{d-1}n^{d-1} + \cdots + a_1 n + a_0,$$

with $d \geq 2$ and $a_i \in \mathbb{Z}$. In the canonical setting we consider prime-generating behavior on $n \geq 0$ and, when convenient, require $f(n) > 0$ for $0 \leq n < L(f)$.

Remark 2.3. The use of $|f(n)|$ in Definition 2.1 allows one to treat polynomials that may take negative values. However, for the main structural results, we will typically work with canonical monic polynomials that are positive on the relevant prime-generating interval.

Definition 2.4 (Genealogical shift)**.** Let $f(n) \in \mathbb{Z}[n]$ be a monic polynomial of degree $d \geq 2$. We define the *forward* and *backward* shifts by

$$S^+(f)(n) := f(n+1), \qquad S^-(f)(n) := f(n-1).$$

Whenever these are interpreted as polynomials in n, they remain monic of degree d with integer coefficients.

Definition 2.5 (Parent and child)**.** Let f and g be canonical monic polynomials of the same degree. We say that g is a *parent* of f and that f is a *child* of g if

$$f(n) = g(n-1)$$

as polynomials in $\mathbb{Z}[n]$. Equivalently, $f = S^-(g)$ and $g = S^+(f)$.

Definition 2.6 (Genealogical tree)**.** Fix a degree $d \geq 2$. Consider the directed graph whose vertices are canonical monic polynomials of degree d and where there is a directed edge

$$g \longrightarrow f$$

if and only if f is a child of g in the sense of Definition 2.5. Each connected component of this graph is called a *genealogical tree*. A polynomial with no parent (in the class under consideration) is called a *root* of its tree.

3 Main theorem

We now state the genealogical theorem. Informally, it asserts that any canonical monic polynomial f with $L(f) > 1$ is the child of a unique canonical monic polynomial p with $L(p) = L(f) - 1$.

Theorem 3.1 (Genealogical Theorem)**.** *Let $f(n) \in \mathbb{Z}[n]$ be a canonical monic polynomial of degree $d \geq 2$ with run length $L := L(f) > 1$. Define*

$$p(n) := f(n+1).$$

Then:

1. $p(n)$ is a canonical monic polynomial of degree d;

2. $L(p) = L - 1$;

3. p is the unique canonical monic polynomial such that $f(n) = p(n-1)$.

In particular, every canonical monic polynomial with $L(f) > 1$ has a unique parent p with $L(p) = L(f) - 1$.

4 Proof of the main theorem

Step 1: p is canonical monic

Since f is monic of degree d, it can be written as

$$f(n) = n^d + a_{d-1}n^{d-1} + \cdots + a_1 n + a_0$$

with $a_i \in \mathbb{Z}$. Substituting $n \mapsto n+1$ gives

$$p(n) = f(n+1) = (n+1)^d + a_{d-1}(n+1)^{d-1} + \cdots + a_1(n+1) + a_0.$$

The leading term of $(n+1)^d$ is n^d with coefficient 1, and all coefficients of $(n+1)^k$ are integers. Therefore $p(n)$ is again a monic polynomial of degree d with integer coefficients, that is, a canonical monic polynomial, proving part (1).

Step 2: lower bound on $L(p)$

Recall that $L = L(f)$ means

$$|f(n)| \text{ is prime for all } n = 0, 1, \ldots, L-1,$$

and that L is maximal with this property.

By definition

$$p(n) = f(n+1).$$

For $n = 0, 1, \ldots, L-2$ we have $n+1 = 1, 2, \ldots, L-1$. By the defining property of L, the values $|f(1)|, |f(2)|, \ldots, |f(L-1)|$ are all prime. Hence

$$|p(n)| = |f(n+1)| \text{ is prime for all } n = 0, 1, \ldots, L-2.$$

It follows that $L(p) \geq L - 1$.

Step 3: upper bound on $L(p)$

Now consider $n = L - 1$. We have

$$p(L-1) = f(L).$$

By maximality of $L = L(f)$, the index L is the first integer for which $|f(L)|$ is not counted as part of the prime run. Formally, either $|f(L)|$ is composite or $f(L)$ fails whatever positivity constraints are imposed in the canonical setting. In any case, $|p(L-1)| = |f(L)|$ does not qualify as a prime value extending the run.

Thus the prime run of $p(n)$ cannot extend beyond $n = L - 2$, and we conclude

$$L(p) \leq L - 1.$$

Together with the lower bound $L(p) \geq L - 1$ from Step 2, this yields

$$L(p) = L - 1,$$

proving part (2).

Step 4: uniqueness of the parent

Suppose that $q(n)$ is a canonical monic polynomial such that

$$f(n) = q(n-1)$$

as polynomials in $\mathbb{Z}[n]$. Replacing n by $n+1$ gives

$$f(n+1) = q(n).$$

Therefore

$$q(n) = f(n+1) = p(n)$$

for all n, which implies $q = p$ identically as polynomials. This proves that p is the *unique* canonical monic polynomial satisfying $f(n) = p(n-1)$, hence the unique genealogical parent of f.

Combining the four steps completes the proof of Theorem 3.1. □

5 Corollaries

Corollary 5.1 (Finite genealogical chains)**.** *Let f be a canonical monic polynomial with run length $L(f) = L > 0$. Then there exists a unique finite chain of canonical monic polynomials*

$$f_1, f_2, \ldots, f_L$$

such that:

1. *$f_L = f$;*
2. *for each $k = 2, \ldots, L$, f_k is the child of f_{k-1}, i.e. $f_k(n) = f_{k-1}(n-1)$;*
3. *for each $k = 1, \ldots, L$, we have $L(f_k) = k$.*

In particular, f_1 is a root with $L(f_1) = 1$, and f sits at depth $L-1$ in its genealogical tree.

Proof. Apply Theorem 3.1 recursively. Starting from $f_L := f$ with $L(f_L) = L$, define f_{L-1} to be its unique parent, so that $L(f_{L-1}) = L - 1$. Continuing in this manner, after $L-1$ steps we obtain f_1 with $L(f_1) = 1$. Uniqueness follows from the uniqueness of the parent at each step. □

Corollary 5.2 (Leaves as structural maxima)**.** *Let f be a canonical monic polynomial with run length $L(f) = L > 1$. If f has no child g with $L(g) = L + 1$ (for example, because $f(-1)$ is composite or non-positive), then f is a leaf of its genealogical tree and represents a structural maximum of L within that tree.*

6 Computational examples

We now illustrate the genealogical framework with two computational case studies: a quartic example with run $L = 38$ and a cubic example with run $L = 31$. In both cases, all primality tests were carried out deterministically for the relevant range of values.

6.1 Quartic example: a monic polynomial with $L = 38$

Consider the monic quartic polynomial

$$Q(n) = n^4 - 74n^3 + 1525n^2 - 5772n + 10159.$$

By exhaustive computation, one verifies that

$$|Q(n)| \text{ is prime for } n = 0, 1, \ldots, 37,$$

while $Q(38)$ is composite. Hence the run length of Q is

$$L(Q) = 38.$$

Within the canonical category of monic quartic polynomials evaluated at $n \geq 0$ and tested on $|Q(n)|$, this appears to be a record run length.

According to Theorem 3.1 and Corollary 5.1, Q belongs to a unique genealogical chain

$$Q_1, Q_2, \ldots, Q_{38},$$

where $Q_{38} = Q$, $Q_{L+1}(n) = Q_L(n-1)$, and $L(Q_L) = L$ for $L = 1, \ldots, 38$.

Equivalently, one can fix the root Q_1 and recover Q_L via

$$Q_L(n) = Q_1(n - (L-1)).$$

In a concrete computational experiment, one can display a table of the form:

$$\begin{aligned} Q_1(n) &= \text{(root with } L = 1), \\ Q_2(n) &= Q_1(n-1), \\ &\vdots \\ Q_{38}(n) &= Q_1(n-37) = Q(n). \end{aligned}$$

For brevity we omit the full expansion of all intermediate Q_L, but this genealogy can be reconstructed exactly from Q_1 by the shift $n \mapsto n - 1$.

6.2 Cubic example: a monic polynomial with $L = 31$

As a second case study, consider the monic cubic polynomial

$$C(n) = n^3 + 54n^2 - 2329n + 20507.$$

Deterministic primality testing shows that

$$|C(n)| \text{ is prime for } n = 0, 1, \ldots, 30,$$

and $C(31)$ is composite, so that

$$L(C) = 31.$$

Again, by Theorem 3.1 the polynomial C sits at the top of a unique genealogical chain

$$C_1, C_2, \ldots, C_{31},$$

where $C_{31} = C$, $C_{L+1}(n) = C_L(n-1)$, and $L(C_L) = L$. Equivalently, for a chosen root C_1 with $L(C_1) = 1$, one has

$$C_L(n) = C_1(n - (L-1)).$$

As in the quartic case, the full list of intermediate polynomials can be recovered algorithmically; the explicit display of all 31 levels is omitted here in the interest of space.

7 Computational methods

We briefly summarize the algorithms used to verify run lengths and explore genealogical trees.

7.1 Run-length verification

Given a monic polynomial $f(n)$ and an upper bound $N_{\max}$, we compute $L(f)$ as follows:

1. Set $n := 0$.
2. Compute $v := f(n)$.
3. If $v \leq 1$, stop and set $L(f) := n$.
4. Test $|v|$ for primality; if it is composite, stop and set $L(f) := n$.
5. Otherwise, increment $n := n + 1$ and repeat, provided $n \leq N_{\max}$.

In practice, $N_{\max}$ is taken strictly larger than the expected run length. For the range of values encountered in our experiments, deterministic primality tests (based on trial division up to $\sqrt{|v|}$ and/or deterministic variants of Miller–Rabin for 64-bit integers) suffice.

7.2 Genealogical exploration

To explore genealogical trees, one can adopt either a top-down or a bottom-up strategy.

In the *top-down* approach, starting from a given polynomial f with known $L(f)$, one repeatedly forms the parent $p(n) = f(n + 1)$ and recomputes $L(p)$ until a root with $L = 1$ is reached.

In the *bottom-up* approach, one first searches for canonical monic polynomials with $L = 1$ in a bounded coefficient domain and then applies the child map

$$g(n) = P(n - 1)$$

whenever the newly introduced value $P(-1)$ is prime and extends the run. This can give rise to tall genealogical "sequoias", with leaves attaining record values of L.

8 Discussion and open problems

The genealogical theorem provides a structural lens through which to view prime-generating monic polynomials. Rather than isolated curiosities, polynomials with long prime runs appear as leaves of trees rooted at $L = 1$ elements. Record examples such as the quartic with $L = 38$ and the cubic with $L = 31$ thus acquire a natural genealogical interpretation as structural maxima within their respective trees.

Several questions arise naturally:

- Can one obtain theoretical upper bounds on $L(f)$ in terms of the degree d and the size of the coefficients?
- What is the distribution of genealogical heights among canonical monic polynomials in fixed degree?
- Are there infinite families of roots whose genealogical trees contain arbitrarily large leaves?
- How does the genealogical structure interact with known obstructions to prime generation (e.g. local congruence constraints)?

We hope that the combination of a simple structural theorem with explicit computational examples may stimulate further investigation into the genealogy of prime-generating polynomials.

References

References

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